

innovate achieve lead

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Laboratory-8: Tasks

Submit at https://forms.gle/LpPMtxTsavZbzwas9

by 6:50 PM

Task 1

- Use the glass.csv dataset, to build a machine learning model using kNN to predict the type of glass. Use 30% of the data for testing.
- Check if there are any missing data, if present then handle them properly. Normalize before using the data to train.
- Print the accuracy score and confusion matrice of the model.
- Plot kNN accuracy score vs k (max k = 15)
- Provide the best set of hyperparameters for the model (i.e. identify the parameter values for which kNN achieved the highest accuracy scores)

Task 2

- For the steel plate dataset available at:

```
pip install ucimlrepo #to install ucimlrep

from ucimlrepo import fetch_ucirepo
steel = fetch_ucirepo(id=198)
X = steel.data.features
y = steel.data.targets

y_integers = np.argmax(y.values, axis=1) # to convert y to a vector

class_names = y.columns.tolist()   # class labels if required
```

- Develop a NN classifier using TensorFlow and Keras, with two hidden layers each having 16 and 32 neurons, ReLU activation and a output layer. Use a train-test split of 70:30. Use appropriate learning rate and number of epochs.
- Plot the confusion matrix of the classifier. Also plot the training and testing accuracy and loss against the number of epochs.
- Compare the accuracy and confusion matrix if the NN model has one hidden layer with 8 neurons.
- Compare the accuracy and confusion matrix of the best NN model with Naive Baye's, kNN, and SVM.